

ELEVEN EQUILATERAL POINTS IN ℓ_1^5 REQUIRE AT LEAST NINETEEN COORDINATE GAPS

ZIJIAN ZENG

ABSTRACT. Kusner’s taxicab conjecture predicts that an equilateral set in ℓ_1^n has at most $2n$ points. The first unresolved case asks whether eleven equilateral points can exist in ℓ_1^5 . We prove a structural obstruction: in any such hypothetical configuration, the sum, over the five coordinates, of the numbers of positive consecutive gaps is at least 19. Equivalently, the five coordinates must assume at least 24 distinct levels in total, with repetitions across coordinates counted separately. The proof converts the taxicab metric to five nested weighted cut chains, uses a Parseval-frame spectral deficit, compresses singleton cuts through rank nine, and excludes the remaining rank-ten case by a binary antichain identity. Exact finite enumeration certifies only the rational partition and endpoint tables. The theorem does not rule out configurations with nineteen or more gaps and therefore does not settle Kusner’s conjecture in dimension five.

1. INTRODUCTION

For a normed space X , write $e(X)$ for the largest cardinality of a set whose pairwise distances are all equal. Kusner’s conjecture for the rectilinear norm asserts

$$e(\ell_1^n) = 2n.$$

The lower bound is witnessed by the cross polytope $\{\pm e_1, \dots, \pm e_n\}$. Bandelt, Chepoi, and Laurent proved the three-dimensional case [1]; Koolen, Laurent, and Schrijver proved the four-dimensional case and formulated the problem in terms of nested cut families [3]. A current account by Ge, Xu, and Zhou states that the $p = 1$ formula is known only for $n \leq 4$ [2]. Thus the first open case is whether $e(\ell_1^5) = 10$.

Our result does not decide that question. It eliminates the first nine support strata above the rank lower bound.

Date: August 29, 2026.

2020 Mathematics Subject Classification. 46B20, 52A21, 05B30.

Key words and phrases. equilateral set, rectilinear space, cut metric, Parseval frame, Kusner conjecture.

Theorem 1.1. *Let $x_1, \dots, x_{11} \in \ell_1^5$ be equilateral. For coordinate j , let q_j be the number of distinct values among $x_{1j}, \dots, x_{11,j}$. Then*

$$\sum_{j=1}^5 (q_j - 1) \geq 19, \quad \text{and hence} \quad \sum_{j=1}^5 q_j \geq 24.$$

The quantity $t = \sum_j (q_j - 1)$ counts positive consecutive gaps after sorting each coordinate. Cut copies from different coordinates are counted separately even when they define the same bipartition. The proof is exact: all displayed finite comparisons are rational or integer-power comparisons.

2. FIVE CHAINS AND A PARSEVAL CUT FRAME

Scale the common distance to one and label the points by [11]. In coordinate j , choose a total order refining the weak order of its values. If $S_{j,k}$ is the prefix before the k -th positive consecutive gap and $g_{j,k} > 0$ is that gap, telescoping gives

$$|x_{rj} - x_{sj}| = \sum_k g_{j,k} \delta_{S_{j,k}}(r, s),$$

where δ_S is the cut metric of $S \mid S^c$. Consequently the unit discrete metric on [11] is a positive weighted sum of cuts from five chains. This is the standard nested-cut reduction [3]; it includes ties because zero gaps are simply omitted.

Put

$$H = I_{11} - \frac{1}{11} J_{11}, \quad v_S = H \mathbf{1}_S.$$

Double centering the cut-metric identity yields

$$(1) \quad \sum_{a=1}^t g_a v_{S_a} v_{S_a}^T = \frac{1}{2} H, \quad g_a > 0.$$

Thus $a_a = \sqrt{2g_a} v_{S_a}$ form a Parseval frame in the ten-dimensional space $V = \mathbf{1}^\perp$, and in particular $t \geq 10$.

2.1. The Naimark chain-length bound.

Lemma 2.1. *If the frame has t columns, then one coordinate contains at most $t - 9$ positive gaps.*

Proof. Let A have the a_a as columns. The complementary projection $Q = I - A^T A$ has rank $t - 10$; write its entries as $\langle y_a, y_b \rangle$ for vectors $y_a \in \mathbb{R}^{t-10}$. Two distinct prefixes from one coordinate are nested, so their centered indicators have positive inner product. Hence the corresponding y_a, y_b have strictly negative inner product.

A family of h nonzero vectors with pairwise strictly negative inner products in $\mathbb{R}^r$ has $h \leq r + 1$. Indeed, a dependence cannot have both positive and negative coefficients: moving the negative terms to the other side and taking the inner product of the two positive combinations would equate a squared norm with a strictly negative number. Therefore the dependence

space has dimension at most one. If a coordinate has at least two gaps, none of its complementary vectors is zero; if it has exactly one gap, the claimed inequality is immediate. Taking $r = t - 10$ proves the result. This explicitly includes the one-gap boundary case. $\square$

3. SPECTRAL DEFICIT OF A CHAIN

Group the frame columns by coordinate. Let A_j be the matrix of the $\ell_j = q_j - 1$ columns in coordinate j , and set

$$G_j = A_j^T A_j, \quad \varepsilon_j = \ell_j - \text{tr } G_j.$$

Since $\sum_j A_j A_j^T = I_V$, every eigenvalue of G_j is at most one, and

$$(2) \quad \varepsilon_j \geq 0, \quad \sum_{j=1}^5 \varepsilon_j = t - 10.$$

Consider one chain with $\ell \geq 2$ supported prefix sizes $1 \leq k_1 < \dots < k_\ell \leq 10$. The correlation matrix R of its normalized centered indicators has adjacent correlations $r_i \in (0, 1)$ and

$$R_{ab} = r_a \cdots r_{b-1} \quad (a < b), \quad \prod_{i=1}^{\ell-1} r_i = \sqrt{\frac{k_1(11 - k_\ell)}{k_\ell(11 - k_1)}} \geq \frac{1}{10}.$$

If $G = DRD \leq I$, with D positive diagonal, Cauchy–Schwarz in the R -inner product gives $\text{tr } G \leq \mathbf{1}^T R^{-1} \mathbf{1}$. Direct expansion of the Markov inverse gives

$$y^T R^{-1} y = y_1^2 + \sum_{i=1}^{\ell-1} \frac{(y_{i+1} - r_i y_i)^2}{1 - r_i^2},$$

and hence

$$\varepsilon_j \geq \sum_{i=1}^{\ell-1} \frac{2r_i}{1 + r_i}.$$

The map $s \mapsto 2e^s/(1 + e^s)$ is strictly convex for $s < 0$. Jensen's inequality therefore yields

$$(3) \quad \varepsilon_j \geq \frac{2(\ell_j - 1)}{10^{1/(\ell_j - 1)} + 1} \quad (\ell_j \geq 2).$$

The following rational lower bounds, proved by raising positive quantities to integer powers, will be used:

$$(4) \quad \begin{array}{c|cccccccccc} \ell & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline c_\ell & 0 & 0 & \frac{2}{11} & \frac{24}{25} & \frac{19}{10} & \frac{23}{8} & \frac{193}{50} & \frac{97}{20} & \frac{146}{25} & \frac{683}{100} \end{array}$$

Every c_ℓ is at most the right side of (3). Their successive increments are

$$0, \frac{2}{11}, \frac{214}{275}, \frac{47}{50}, \frac{39}{40}, \frac{197}{200}, \frac{99}{100}, \frac{99}{100}, \frac{99}{100},$$

so the sequence is discretely convex. In particular (2) and a one-unit exchange first exclude $t = 10, \dots, 14$. For example the minimum costs at $t = 12, 13, 14$ are respectively

$$\frac{678}{275} > 2, \quad \frac{892}{275} > 3, \quad \frac{1106}{275} > 4.$$

3.1. Endpoint penalties. A chain is *regular* if its endpoint prefix sizes are $(1, 10)$; then it supplies two singleton cut copies. If it loses $e \in \{0, 1, 2\}$ of these endpoints, its endpoint correlation θ satisfies

$$\theta^2 \geq \frac{1}{45} \quad (e \geq 1), \quad \theta^2 \geq \frac{4}{81} \quad (e = 2).$$

Substitution in the same Jensen bound gives the safe table

ℓ	general	$e \geq 1$	$e = 2$
2	$\frac{2}{11}$	$> \frac{1}{4}$	$\geq \frac{4}{11}$
3	$> \frac{24}{25}$	$> \frac{111}{100}$	$> \frac{5}{4}$
4	$> \frac{19}{10}$	$> \frac{207}{100}$	$> \frac{9}{4}$
5	$> \frac{23}{8}$	$> \frac{153}{50}$	$> \frac{13}{4}$
6	$> \frac{193}{50}$	$> \frac{81}{20}$	$> \frac{17}{4}$
7	$> \frac{50}{97}$	$> \frac{101}{20}$	$> \frac{21}{4}$

Strict and non-strict signs in this table are retained in every sum below. For instance, if $L < 2(\ell - 1)$, the comparison with the Jensen bound is equivalent to comparing θ^2 with $(L/(2(\ell - 1) - L))^{2(\ell - 1)}$; thus no floating-point radical comparison is involved.

4. SINGLETON COMPRESSION THROUGH RANK NINE

Call a cut copy singleton if one side has one label. Let R be the number of non-singleton copies. Put $s_r = He_r$, let α_r be the total weight of singleton copies labelled by r , and set

$$c_r = \frac{1}{2} - \alpha_r, \quad Z = \{r : c_r = 0\}.$$

Subtracting singleton terms from (1) gives

$$(6) \quad C = H \operatorname{diag}(c) H = \sum_{a=1}^R g_a v_{S_a} v_{S_a}^T \succeq 0.$$

Lemma 4.1. *No decomposition (1) has $1 \leq R \leq 9$.*

Proof. On V , a kernel vector of the left side of (6) satisfies

$$c_r u_r = \lambda \quad (1 \leq r \leq 11)$$

for a constant λ . If $Z \neq \emptyset$, then $\lambda = 0$ and

$$\ker(C|_V) = \{u : \operatorname{supp} u \subseteq Z, \sum_{r \in Z} u_r = 0\}.$$

This formula also handles $|Z| = 1$: the kernel is zero and $\text{rank } C = 10 > R$. In every remaining zero-set case, positivity of every g_a makes each non-singleton cut indicator constant on Z . If $Z = [11]$, this constancy would make every remaining cut trivial, contrary to $R \geq 1$; hence $W = [11] \setminus Z$ is nonempty. Orient the cut to a nonempty subset U_a of W . The vectors $(s_w)_{w \in W}$ are independent, so a left inverse applied to (6) gives

$$\text{diag}(c_w : w \in W) = \sum_{a=1}^R g_a \mathbf{1}_{U_a} \mathbf{1}_{U_a}^T.$$

All g_a are strictly positive. The zero off-diagonal entries force every U_a to contain at most one label; being nonempty, it is a singleton, a contradiction.

It remains to consider $Z = \emptyset$. The displayed kernel equations give kernel dimension at most one, so rank at least nine. Thus only $R = 9$ is possible, with rank exactly nine and $\sum_r 1/c_r = 0$. Positive semidefiniteness on V permits at most one negative c_r , while the reciprocal identity prevents all of them from being positive. Hence $(1/c_r)_r$ has exactly one negative and ten positive entries, has total sum zero, and has no nonempty proper zero-sum subset. Every remaining cut vector must be orthogonal to this kernel vector, contradicting that fact. This proves the lemma. $\square$

5. THE FINITE STRATA $t = 15, 16, 17, 18$

Discrete convexity of (4), the exact budget (2), and the endpoint penalties (5) leave the following sorted gap-count patterns:

t	surviving patterns
15	(3, 3, 3, 3, 3), (4, 3, 3, 3, 2)
16	(4, 3, 3, 3, 3), (4, 4, 3, 3, 2), (5, 3, 3, 3, 2)
17	(4, 4, 3, 3, 3), (5, 3, 3, 3, 3), (4, 4, 4, 3, 2), (5, 4, 3, 3, 2), (6, 3, 3, 3, 2)
18	(4, 4, 4, 3, 3), (5, 4, 3, 3, 3), (6, 3, 3, 3, 3), (4, 4, 4, 4, 2), (5, 4, 4, 3, 2), (5, 5, 3, 3, 2), (6, 4, 3, 3, 2), (7, 3, 3, 3, 2).

This list is exhaustive: it checks all integer partitions satisfying $\ell_j \leq t - 9$, the latter being Lemma 2.1. For $t = 18$, for example, there are 88 candidate partitions before the cost test, and exactly the eight displayed survive.

Assign each chain an endpoint loss $e \in \{0, 1, 2\}$ and sum the appropriate entries of (5). Exhausting the 3^5 assignments per pattern proves:

t	retained singleton endpoints	resulting bound on R
15	9 or 10	$R = 6$ or 5
16	at least 8	$1 \leq R \leq 8$
17	at least 8	$1 \leq R \leq 9$
18	at least 8	$8 \leq R \leq 10$.

For $t = 15$, the deficit inequalities say that the all-triple pattern is either fully regular or has exactly one chain with exactly one singleton endpoint;

the other pattern is fully regular. These are finite rational comparisons, not numerical sampling. Lemma 4.1 therefore excludes $t = 15, 16, 17$. At $t = 18$, it excludes $R \leq 9$; only $R = 10$ remains.

6. THE RANK-TEN ANTICHAIN OBSTRUCTION

Assume $R = 10$ in (6). Fix a reference label p and put $W = [11] \setminus \{p\}$. Orient every non-singleton cut to the side $U_a^{(p)}$ not containing p . Since both sides are non-singleton,

$$(9) \quad 2 \leq |U_a^{(p)}| \leq 9.$$

The ten vectors $(s_r)_{r \in W}$ form a basis of V , with $s_p = -\sum_{r \in W} s_r$. Let B_p be the binary matrix whose columns are the $\mathbf{1}_{U_a^{(p)}}$, let $G = \text{diag}(g_1, \dots, g_{10})$, and $D_p = \text{diag}(c_r : r \in W)$. Expressing (6) in this basis gives

$$(10) \quad D_p + c_p J = B_p G B_p^T.$$

Lemma 6.1. *For fixed p , the ten oriented blocks $U_a^{(p)}$ intersect pairwise and no one contains another.*

Proof. For distinct $i, j \in W$, the off-diagonal entry of (10) is

$$c_p = \sum_{a: i, j \in U_a^{(p)}} g_a \geq 0.$$

If $c_p = 0$, strict positivity of all g_a would make every column of B_p have at most one 1, contrary to (9). Thus $c_p > 0$. Since p was arbitrary, every $c_r > 0$. Therefore the left side of (10) is positive definite and B_p is invertible.

Suppress p , and define

$$s = \sum_{r \in W} \frac{1}{c_r}, \quad \lambda = \frac{c_p}{1 + c_p s}.$$

Then $0 < \lambda s < 1$. Sherman–Morrison gives

$$(D + c_p J)^{-1} = D^{-1} - \lambda D^{-1} J D^{-1}.$$

Because $G^{-1} = B^T (D + c_p J)^{-1} B$ is diagonal, its off-diagonal entries give, for distinct cut copies,

$$\sum_{r \in U_a \cap U_b} \frac{1}{c_r} = \lambda \left(\sum_{r \in U_a} \frac{1}{c_r} \right) \left(\sum_{r \in U_b} \frac{1}{c_r} \right).$$

The right side is positive, so the blocks intersect. If $U_a \subseteq U_b$, cancellation of the positive sum over U_a would give

$$1 = \lambda \sum_{r \in U_b} \frac{1}{c_r} \leq \lambda s < 1,$$

a contradiction. This also rules out repeated blocks. Interchanging a, b excludes the reverse containment. $\square$

Two prefix cuts from one coordinate are nested. After orientation away from p , they are either nested or disjoint: if p lies in neither or both, the oriented sides remain nested, while if p lies between the two prefixes, the oriented sides are disjoint. Lemma 6.1 therefore permits at most one non-singleton copy per coordinate. Five coordinates can supply at most five such copies, contradicting $R = 10$. This eliminates $t = 18$, and Theorem 1.1 follows.

7. EXACT COMPUTATIONAL CHECKS AND A FINITE-BOX RESULT

The proof above is human mathematics. Exact checkers independently reconstruct the finite partition lists, rational endpoint tables, and integer-power comparisons from serialized data. The decisive verifiers use explicit conditions followed by exceptions or nonzero exits; none uses Python `assert` for a mathematical decision. Genuine inputs and tampered inputs were run from outside the project under the four modes `python`, `python -O`, `python -I`, and `python -O -I`. The fail-closed matrix contains 48 expected outcomes for the three foundational verifiers, and the $t = 18$ endpoint manifest contains 60 expected outcomes. All matched their expected PASS/FAIL disposition.

For comparison, a separate exhaustive certificate establishes only the following finite statement.

Proposition 7.1. *For every integer distance D , an equilateral subset of $\{0, 1, 2, 3, 4\}^5$ in the ℓ_1 metric has at most ten points. The maximum over all D is ten.*

The verifier reconstructs all $5^5 = 3125$ vertices and all distance graphs. Odd distances use the parity bound, and even distances use an independent fixed-target clique search with a valid coloring upper bound. This proposition is a finite-box theorem only: it does not imply $e(\ell_1^5) \leq 10$ for arbitrary real or integer coordinates.

All reproducibility scripts require Python 3.10 or newer because the finite-box verifier uses `int.bit_count`. The release manifest records exact SHA-256 hashes. No proof assistant was used.

8. LIMITATIONS AND DISCLOSURE

The theorem excludes only $t \leq 18$. It neither constructs eleven equilateral points nor excludes the infinitely many order types with $t \geq 19$. In particular, Kusner’s original $n = 5$ problem remains open.

Generative AI assisted with literature discovery, proof exploration, exact arithmetic checks, code generation, and manuscript drafting. Every retained claim was reduced to a human proof or a serialized exact finite certificate. The final proof chain and fail-closed execution semantics received independent adversarial audits.

REFERENCES

1. Hans-Jürgen Bandelt, Victor Chepoi, and Monique Laurent, *Embedding into rectilinear spaces*, Discrete & Computational Geometry **19** (1998), no. 4, 595–604.
2. Hong-Jun Ge, Zixiang Xu, and Yang Zhou, *Kusner’s conjecture: Exact values and linear bounds*, 2026.
3. Jack Koolen, Monique Laurent, and Alexander Schrijver, *Equilateral dimension of the rectilinear space*, Designs, Codes and Cryptography **21** (2000), no. 1–3, 149–164.

SCHOOL OF ARTIFICIAL INTELLIGENCE AND AUTOMATION, CHINA UNIVERSITY OF
GEOSCIENCES; INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNI-
VERSITY

Email address: 1002266693@ucsiuniversity.edu.my